

Freeze-Omni: A Smart and Low Latency Speech-to-speech Dialogue Model with Frozen LLM

Deep Learning Paper Review Study Group
Team: Fundamental
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Problem Statement

For interaction with LLM in Speech Modality,

Traditional method :

“Use a cascaded approach of ASR + LLM + TTS”

Limitation :

- High engineering complexity
- High interaction latency.

New method:

*“By the parameters of the LLM are more or less fine-tuned,
Aligning the LLM with the speech modality”*

Limitation :

- The forgetting problem to the LLM, resulting in a negative impact on its intelligence
- an obvious gap in performance between spoken question-answering and text-modality question-answering

Freeze-Omni suggests:

“Achieving speech modality alignment while the LLM is frozen throughout the training process, and obtaining low latency speech dialogue capabilities while keeping the intelligence of the backbone LLM.”

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Freeze-Omni: A Smart and Low Latency Speech-to-speech Dialogue Model with Frozen LLM

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<https://freeze-omni.github.io/>

Abstract

The rapid development of large language models has brought many new smart applications, especially the excellent multimodal human-computer interaction in GPT-4o has brought impressive experience to users. In this background, researchers have proposed many multimodal LLMs that can achieve speech-to-speech dialogue recently. In this paper, we propose a speech-text-music multimodal LLM architecture called Freeze-Omni, which can achieve speech-to-speech, speech-to-text, and text-to-speech interaction. Freeze-Omni connects the LLM with keeping the LLM frozen throughout the training. We designed 3-stage training strategies both for the modeling of speech input and output, enabling Freeze-Omni to obtain speech-to-speech interaction with a training cost of only 1.2M tokens. Freeze-Omni achieves **60,000 multi-round text Q&A data on 8 GPUs**. Moreover, we can effectively ensure that the intelligence of the Freeze-Omni in the speech modality is at the same level compared with that in the text modality of its backbone LLM, while the end-to-end latency of the spoken response achieves a low level. In addition, Freeze-Omni achieves a rating of 4.5 on the user satisfaction of the dialogue training, making Freeze-Omni have a more natural style of dialogue ability between the users. Freeze-Omni mainly provides a possibility for researchers to conduct multimodal LLM under the condition of a frozen LLM, avoiding various impacts caused by the catastrophic forgetting of LLM caused by fewer data and training resources.

1 Introduction

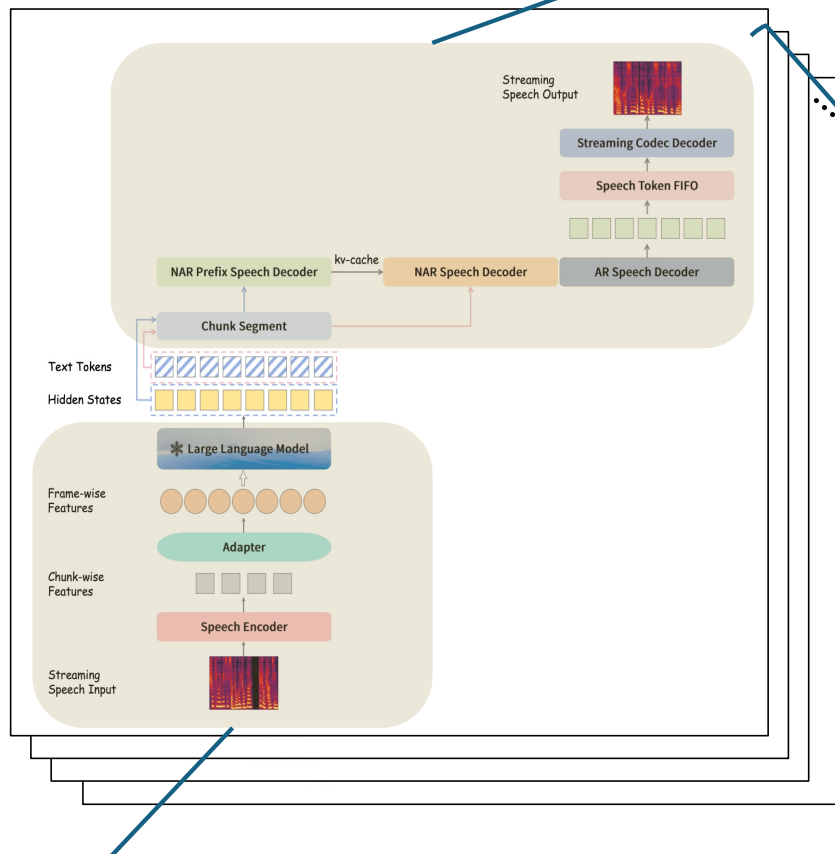
In recent years, the development of large language models has been extremely rapid. A series of large language models represented by the GPT series [10, 11] of OpenAI has demonstrated extraordinary capabilities. As speech interaction is one of the most natural forms of human-computer interaction, combining speech interaction with output with an LLM can bring about extraordinary benefits to users. The traditional method is to use cascaded approach of ASR + LLM + TTS to achieve the interaction with LLM in speech modality. However, this approach often leads to a relatively high engineering complexity and a considerable interaction latency. Nevertheless, GPT4o [18] has changed this situation, it provides an end-to-end speech interaction mode which has significantly improved the user experience, triggering a research boom among researchers regarding multimodal LLMs for speech-to-speech interaction.

In the field of general LLMs, many public models such as Llama 3.2 [8], Qwen2.5 [21], Mixtral [14], etc. have provided very good opportunities for researchers to develop downstream tasks on them. Therefore, in the research field of multimodal LLMs for speech-to-speech, works such as Mini-Omni2 [24], LLaMA-Omni [9], and Moshi [7] have provided excellent references for researchers.

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Overview

Modeling of Speech Output



Design for Duplex Dialogue

Modeling of Speech Input

Overview

Modeling of Speech Input

1) The alignment between the speech input to text output,



Modeling of Speech Output

2) The alignment between the text input to speech output



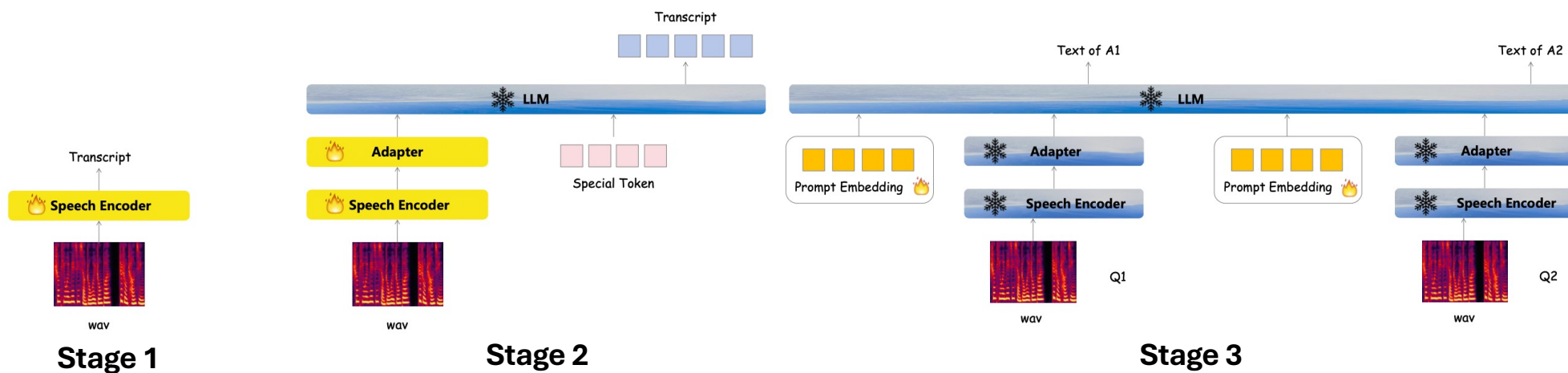
Design for Duplex Dialogue

3) By connecting these two components with the LLM



The Ability of Speech Input to Speech Output

Modeling of Speech Input

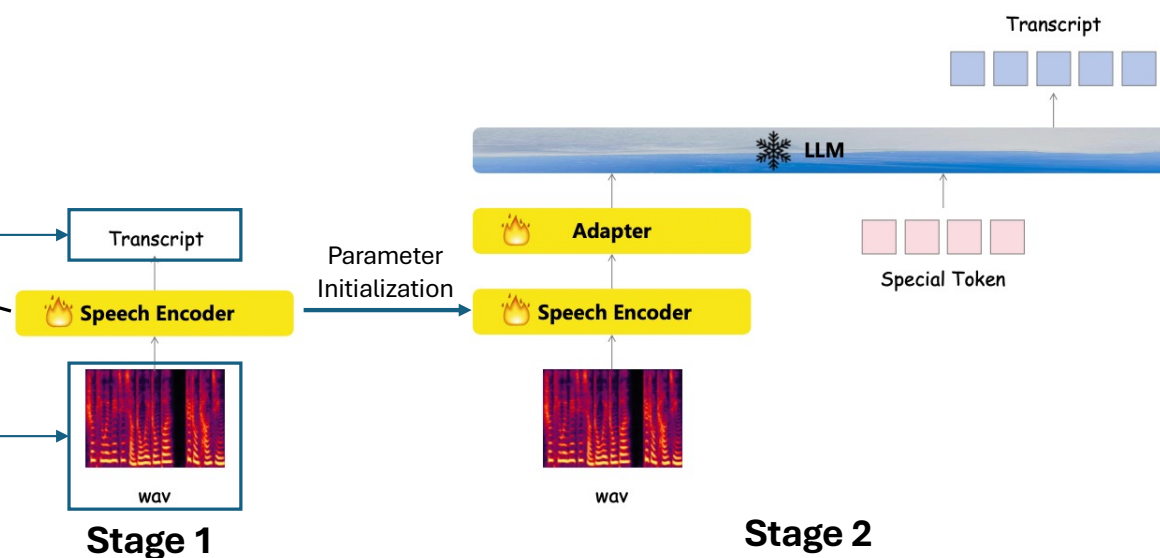


Modeling of Speech Input

Dataset : 110,000h internal speech text paired ASR data including both Chinese and English

Utilizes a chunk-wise streaming speech encoder to transform the input speech features into a high-dimensional representation.

- A multi-layer convolution with 4-times down sampling and 24 layers of transformers with a hidden size of 1024.
- # of Parameter : 350M

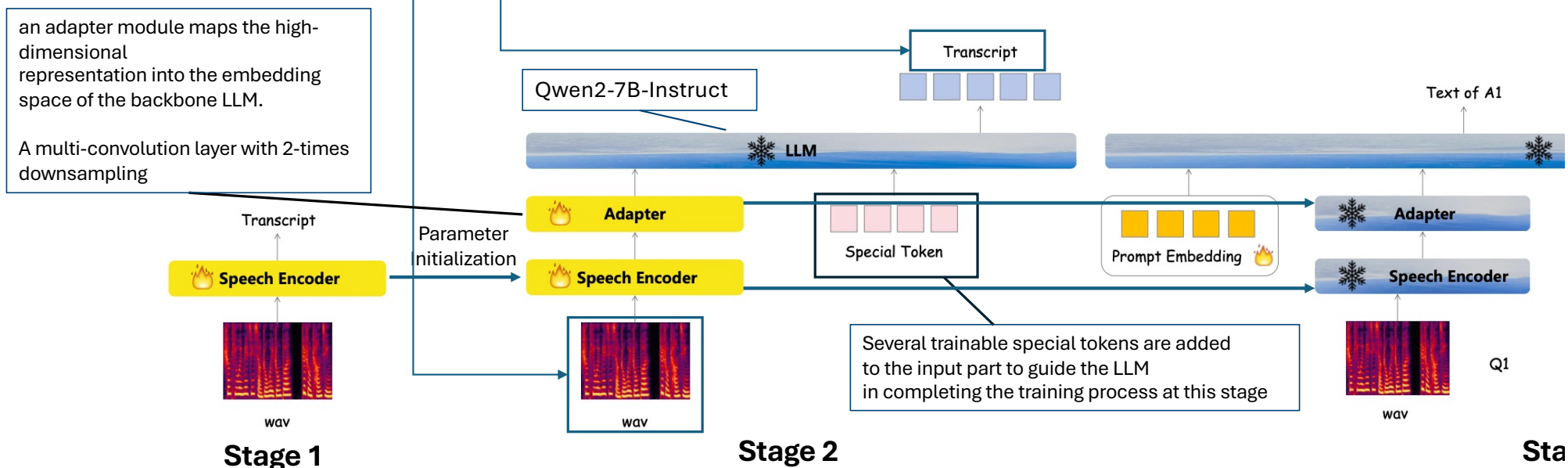


Training :

- Loss Function: CTC
- Optimizer : Adamw [16] optimizer with a warm-up learning rate scheduler
- Learning Rate:
 - Stage 1: 2e-4

Modeling of Speech Input

Dataset : 110,000h internal speech-text paired ASR data including both Chinese and English



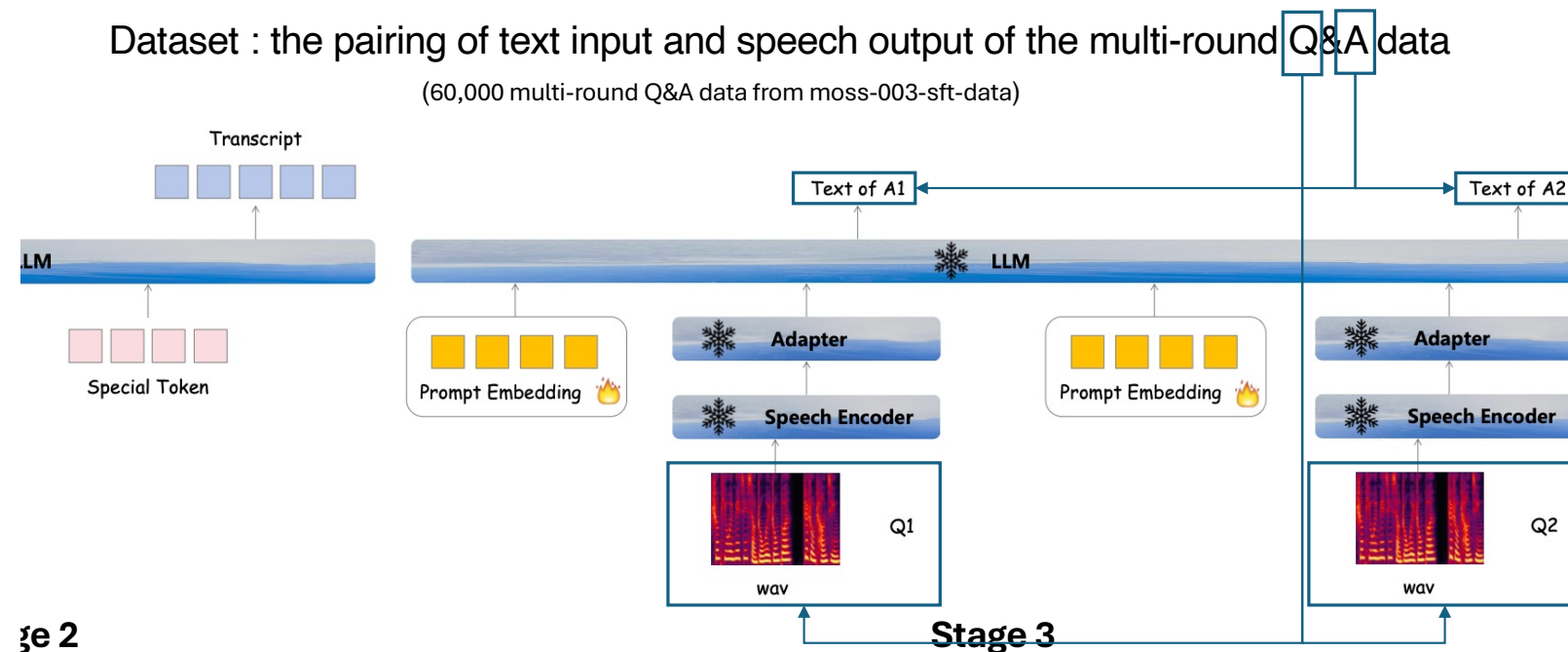
Training :

- Loss Function: CTC
- Optimizer : Adamw [16] optimizer with a warm-up learning rate scheduler
- Learning Rate:
 - Stage 1: 2e-4
 - Stage 2: 1e-4

Modeling of Speech Input

Dataset : the pairing of text input and speech output of the multi-round Q&A data

(60,000 multi-round Q&A data from moss-003-sft-data)



ge 2

Training :

- Loss Function: CTC
- Optimizer : Adamw [16] optimizer with a warm-up learning rate scheduler
- Learning Rate:
 - Stage 1: $2e-4$
 - Stage 2: $1e-4$
 - Stage 3: $6e-4$

Results on Speech Input

Table 1: The ASR performance of the model corresponding to stage 2 in the modeling of speech input, where {aishell-1 [4], test_net [26], test_meeting [26]} are Mandarin evaluation sets, measured in CER (%), while {dev-clean, dev-other, test-clean, test-other} [19] are English evaluation sets, measured in WER (%).

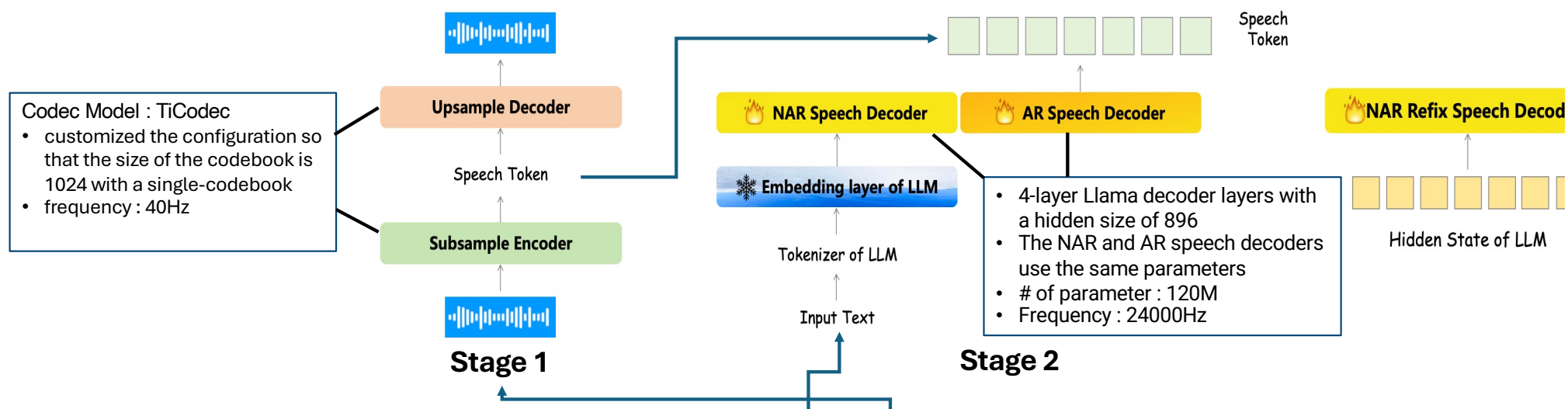
Model	aishell-1	test_net	test_meeting	dev-clean	dev-other	test-clean	test-other
Wav2vec2-base [2]	-	-	-	6.0	13.4	-	-
Mini-Omni2 [24]	-	-	-	4.8	9.8	4.7	9.4
Freeze-Omni							
+ <i>chunk</i> = ∞	2.15	8.57	10.09	3.29	7.4	3.24	7.68
+ <i>chunk</i> = 4	2.79	12.6	14.2	4.16	10.21	4.05	10.48
+ w/o dynamic	2.48	11.8	13.46	4.03	9.45	3.82	9.79

Dynamic Chunk Training enables the model **to handle both streaming and offline conditions** by training with variable chunk sizes.

Although a fixed chunk (e.g., 4) yields slightly lower error rates without dynamics, the **dynamic approach offers far better generalization and flexibility**.

Modeling of Speech Output

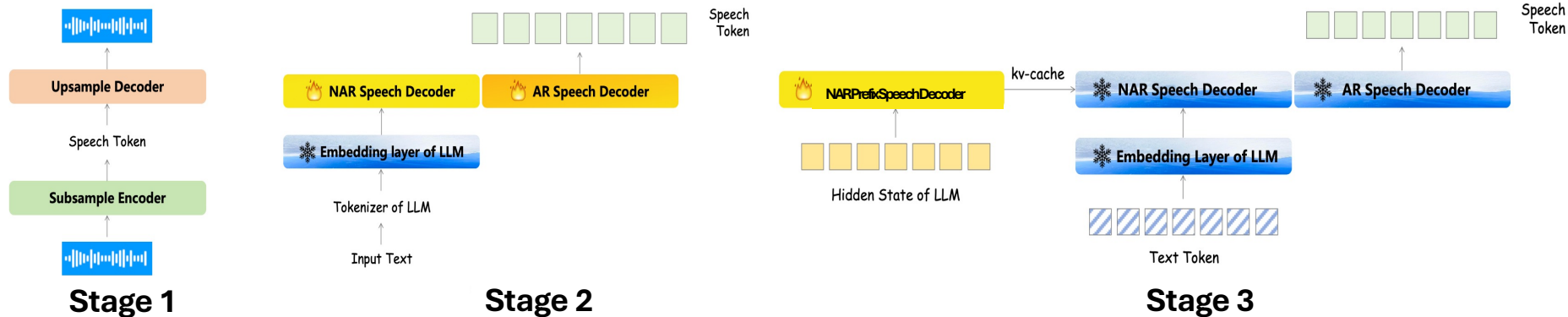
Dataset : 3,000h of text-speech paired data generated by a zero-shot TTS system



Training :

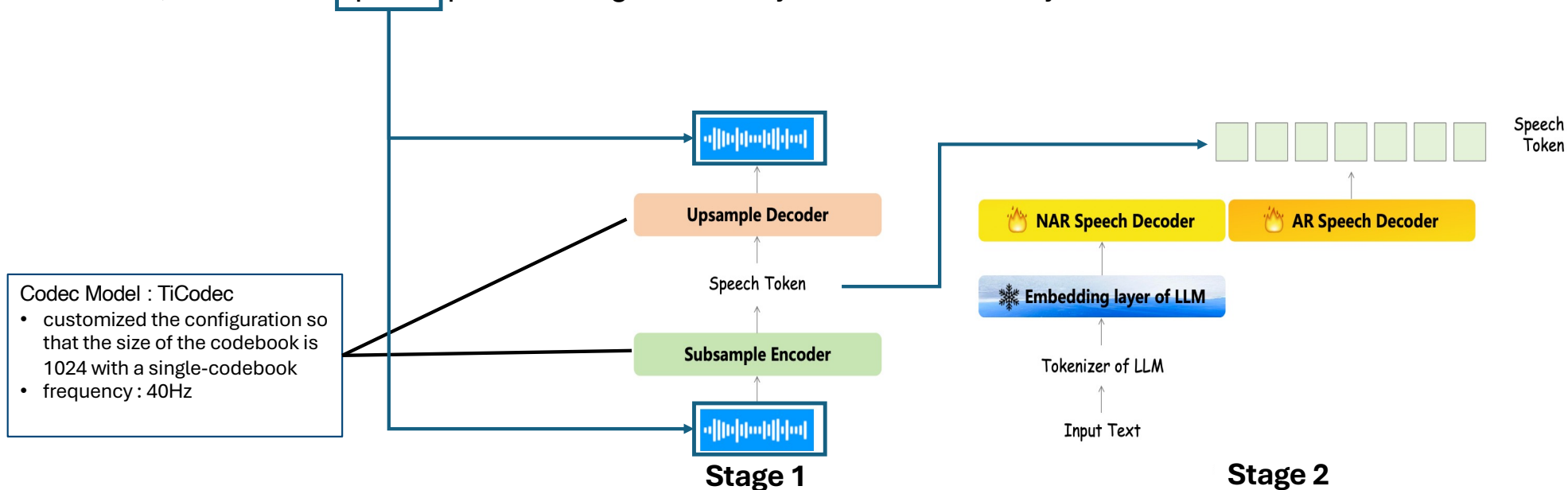
- Loss Function: CTC
- Optimizer : Adamw [16] optimizer with a warm-up learning rate scheduler
- Learning Rate:
 - Stage 1: Same as TiCodec
 - Stage 2: 5e-5
 - Stage 3: 5e-5

Modeling of Speech Output



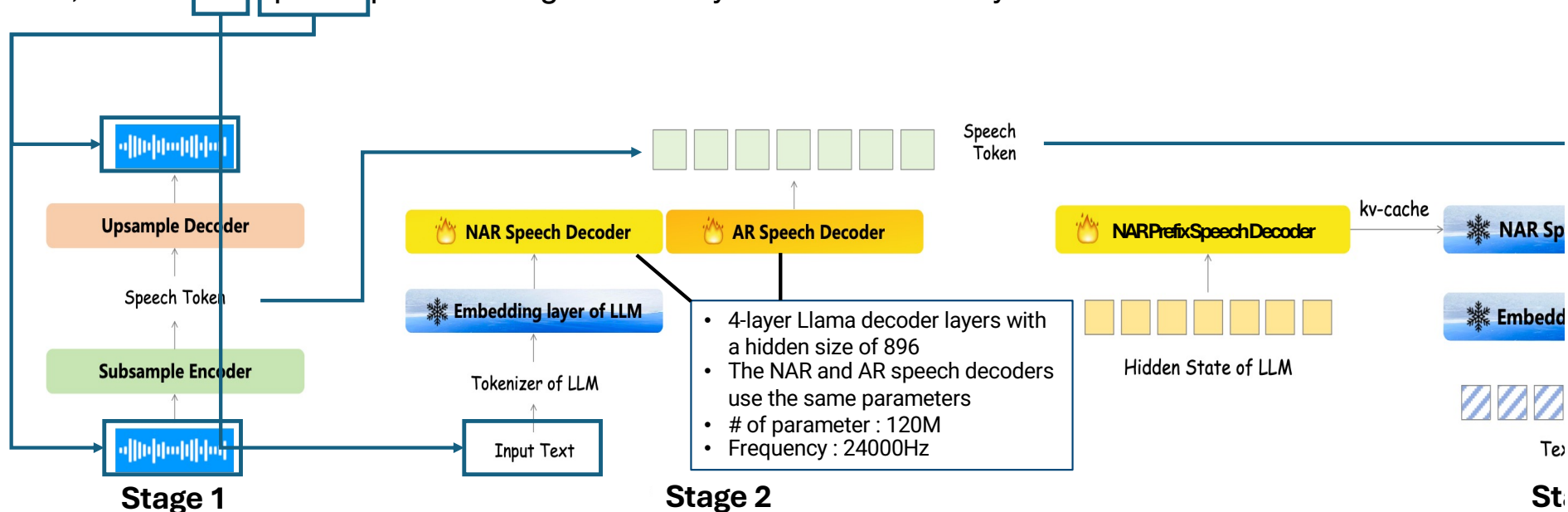
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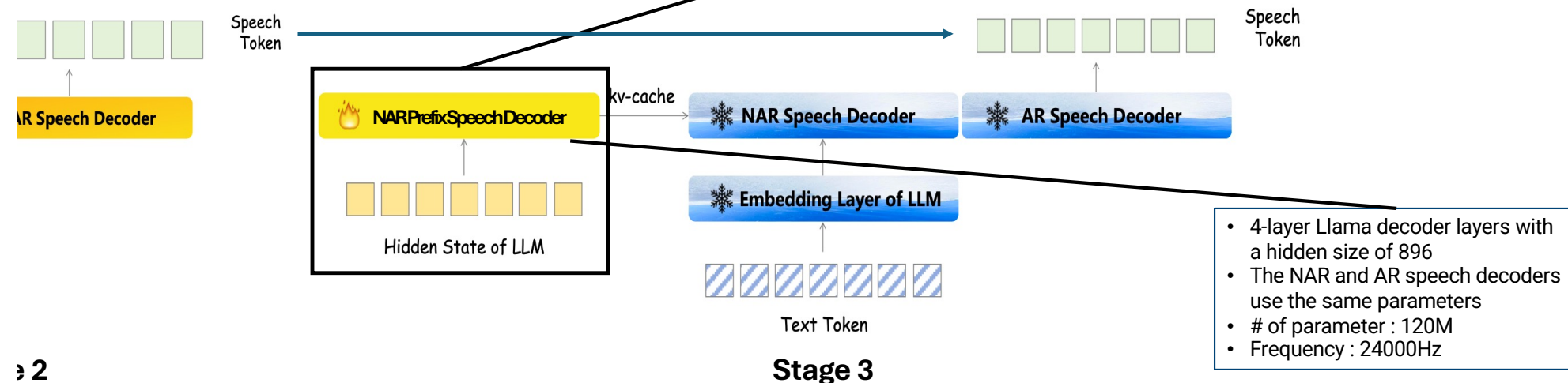
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Modeling of Speech Output

Dataset : the pairing of text input and speech output of the multi-round Q&A data

(60,000 multi-round Q&A data from moss-003-sft-data)

lies in closely coupling the speech decoder with the output of the LLM to reduce the occurrence of bad cases



Training :

- Loss Function: CTC
- Optimizer : Adamw [16] optimizer with a warm-up learning rate scheduler
- Learning Rate:
 - Stage 1: Same as TiCodec
 - Stage 2: 5e-5
 - Stage 3: 5e-5

Results on speech output

Experimental Setup

- 1,000 utterances were randomly sampled, using **text tokens and hidden states** from the LLM as inputs to the speech decoder.
- The generated speech was evaluated by **ASR accuracy (CER%)** using *paraformer-zh*, under different **top-k decoding** settings.
- Two models were compared: **Speech Decoder w/o Prefix** (stage 2) and **Speech Decoder with Prefix NAR** (stage 3).

Table 2: The CER(%) of the speech decoder on 1,000 evaluation utterances under different *top-k*.

Method	<i>top-k</i>				
	1	2	3	4	5
Speech Decoder w/o Prefix	5.27	4.64	4.76	4.66	5.03
Speech Decoder	3.9	3.65	3.53	3.62	3.71

Prefix NAR Decoder **improves alignment** with the LLM and **reduces CER**, showing **better robustness** across all **top-k** values.

Design for duplex dialogue

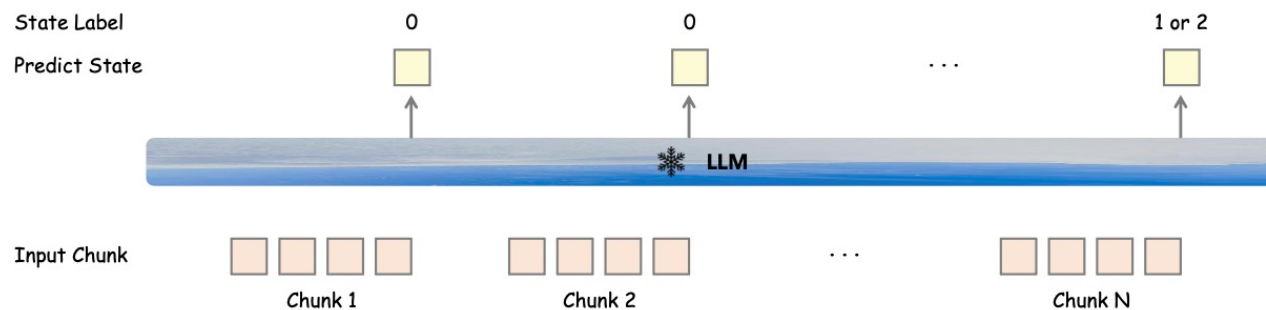
use multi-task for chunk-level state prediction

State 0 : the current LLM can continue to receive speech

State 1 : the LLM can interrupt the user and perform the generate stage

State 2 : there is no need to interrupt the user

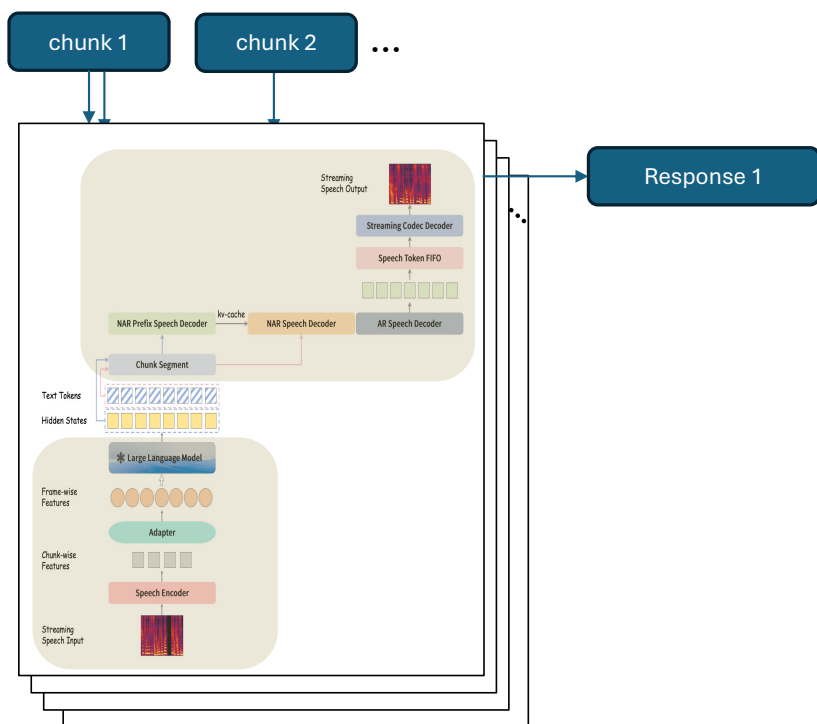
} Stop sending speech streams to Freeze-Omni and reset the VAD module.



Using a multi-task method to optimize the cross-entropy loss of both the state classification layer and the LLM.

Design for duplex dialogue

"model as a server" strategy for the speech-to-speech dialogue system



- Started several models simultaneously
- A user's VAD was triggered, the speech would be sent to the server in the form of chunks
- Server would be responsible for scheduling which idle model should respond to the current chunk
- Separated all the kv-cache and CNN cache of the speech encoder and LLM

“Any model in the server could respond to any chunk of any user”

Results on spoken question answering

Table 3: The accuracy (%) of different models in question answering on three sets. The models in the first four rows all use speech as input, while the models in the last two rows use text as input. The backbone LLM of Freeze-Omni is Qwen2-7B-Instruct, and the backbone LLM of Moshi is Helium. Both Freeze-Omni and Qwen2-7B-Instruct use greedy search in the generate stage with zero-shot, and the accuracy is calculated using the output text. Except for Freeze-Omni and Qwen2-7B-Instruct, previous evaluation results are derived from corresponding references.

Model	Modality	Web Q.	LlaMA Q.	Audio Trivia QA
SpeechGPT(7B) [27]	Audio&Text	6.5	21.6	14.8
Spectron(1B) [17]	Audio&Text	6.1	22.9	-
Moshi(7B) [7]	Audio&Text	26.6	62.3	22.8
Freeze-Omni(7B)	Audio&Text	44.73	72	53.88
Helium [7]	Text Only	32.3	75	56.4
Qwen2-7B-Instruct	Text Only	45.13	77.67	63.93

- **Freeze-Omni** achieves **much higher accuracy** than other speech-based models.
- Its performance is **close to text-only LLMs (Qwen2-7B-Instruct)**, showing strong alignment between **speech and text understanding**.
- Demonstrates that **Freeze-Omni preserves reasoning ability even from spoken input**.

Analysis on end-to-end latency

- **Statistical latency:** Time from LLM interruption to the first generated speech PCM chunk (automatically measurable).
- **Non-statistical latency:** Real-time delay from the end of user speech to LLM's response start (manually measured).

Table 4: Detailed information of statistical latency. Among them, 50% represents the median, and 90% represents the percentile at 90. The unit of the results in the table is (ms). All results are completed using pytorch with bfloat16 inference on a single NVIDIA A100 GPU.

Latency description	Avg.	50%	90%
LLM interrupted → LLM generate first text token chunk	478	468	750
First text token chunk → Prefill of speech decoder	15	15	17
Prefill of speech decoder → Generate first speech token chunk	237	235	252
First speech token Chunk → Decode first PCM hunk	11	11	13
Total	745	753	1020

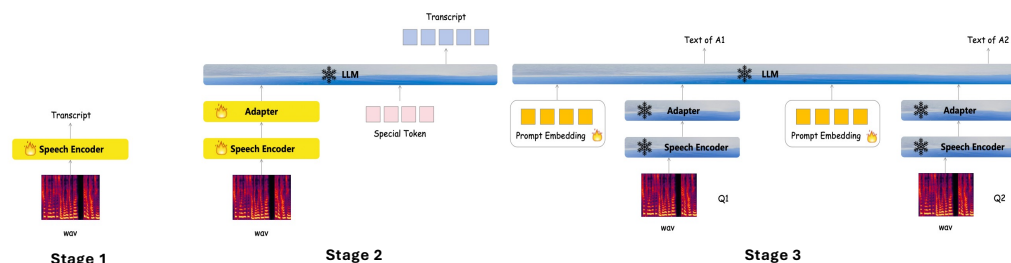
- **Main delay** occurs before the first text token generation (~0.5 s).
- **Speech decoding is extremely fast** (tens of ms).
- Overall, **Freeze-Omni achieves real-time conversational speed (~1.2 s)**, practical for interactive speech dialogue.

Conclusion

- **Freeze-Omni:**
a **text-audio multimodal LLM** enabling **low-latency speech-to-speech dialogue** without fine-tuning the LLM.
- Achieves **strong performance** across multiple evaluation tasks while keeping the LLM **frozen**.

Provide pathway developing strong performance speech modules to next research

Speech Input



Speech Output

